

Lucia Brown – 22336688 – Abandoned & Removed Objects Classification

This is a report for the classification of abandoned and removed objects in OpenCV. The objective of this project is to detect abandoned and removed objects in static camera videos. The approach uses adaptive background modelling, motion detection, and persistence analysis.

Solution Overview:

- The program defines several parameters dealing with thresholds for:
 - Motion detection,
 - How “different” frames should be to be considered significantly different,
 - How long detected changes should last to be considered persistent, etc.

Development Stage 1 - Step 1 – Detect Motion:

- All motion is detected using a Gaussian Mixture Model (GMM) background subtractor, producing a binary video of all movements, inclusive of noise.
- The foreground (motion) is dilated to better detect small abandoned/removed objects.
- The mixture of Gaussians was chosen over a static background or median background model, as the background should evolve over time to account for videos where there are both abandonments and removals of the same object. GMM is adaptive, and less computationally intense than a median model

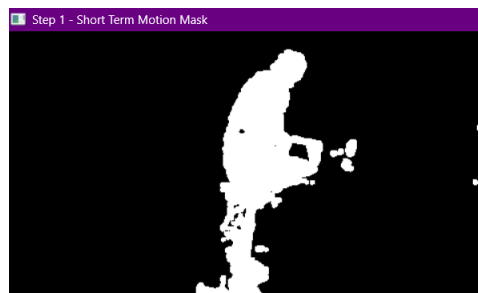


Fig 1.1 – DEVELOPMENT: Binary Output of All Motion

Development Stage 1 - Step 2 – Define a Stable Background Model:

- A stable background in the scene is defined by removing motion pixels below the MOTION_DETECTION_THRESHOLD, a measure of if 1.5% of pixels are moving then there is significant/non-noise movement in the scene.
- A Gaussian blur is used to reduce the noise of the stable background.
- This step produces a smooth, stable background; to be used in comparison against changes.

Development Stage 1 - Step 3 – Accumulate Frames of the Stable Background:

- Once the stable background has accumulated a significant number of frames, specified by LONG_TERM_CHANGE_WINDOW, it is defined as the long-term background.
- For some videos, a true stable background cannot be found due to many moving pixels in the beginning. These videos have a “forced” stable background taken after 1 second of the video. This is to prevent the video from being skipped in the processing.
- “Forced” backgrounds still maintain a level of updating throughout the video, such that multiple detections can be made within one video.
- The “forced” background model uses a median of the first 30 and last 30 frames of the video for basing decisions around detection.
- The median background model was chosen over GMM for this situation, as it has been proven by the threshold not being reached that pixels are updating too continuously, it can be assumed there is no reasonable period of stillness to determine a GMM background.
- A drawback of using a median model for these situations is that movement from the first/last N pixels are being assumed background, which generates false positives/negatives in detection systems. I prefer this outcome over the alternative of only using GMM, which in development, outright skipped over videos where the long term-window was not achieved.

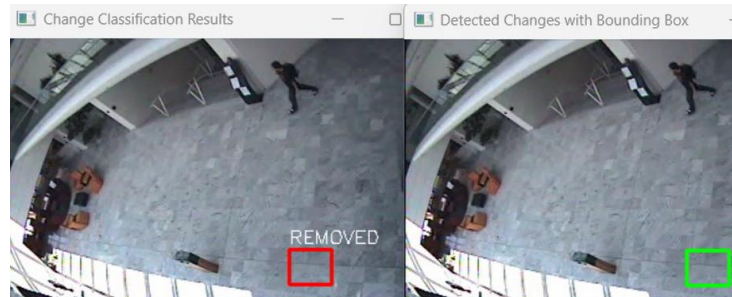


Fig 2.1 – RESULT: A False Positive from Median Background Analysis

Development Stage 1 - Step 4 – Get Difference from the Long-Term Background:

- Pixels differing from the long-term background are identified by obtaining an absolute difference between the background and the current frame.
- This difference is converted to greyscale and then to binary video of significant changes using the DIFFERENCE_FRAME_THRESHOLD defined.
- This step outputs the binary mask of pixels which are significantly different from the long-term background.

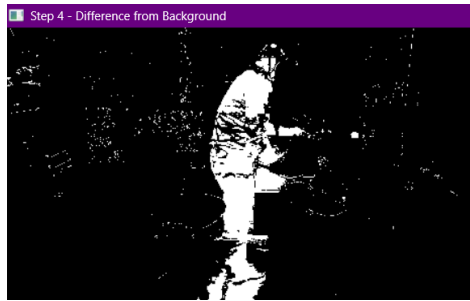


Fig 1.2 - DEVELOPMENT: Binary Output of Significant Movement

Development Stage 1 - Step 5 – Separate Out Transient Motion:

- Bitwise AND of Step 4’s mask and Step 1’s short-term motion is used to separate transient motion from long-term changes, leaving only the long-term changes.
- This binary mask of the long-term changes is cleaned with opening and closing.

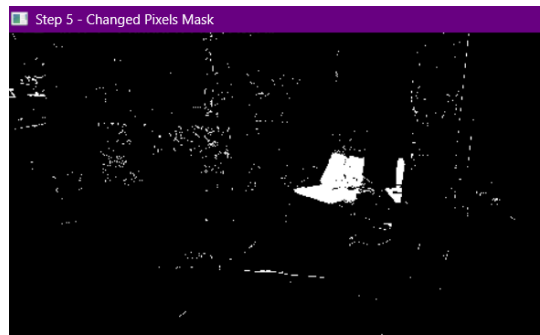


Fig 1.3 – DEVELOPMENT: Binary Output of Removed Transient Motion

Development Stage 1 - Step 6 – Count Changing Pixels/Persistence Analysis:

- Attempting to remove other motion which is classified as long-term but may not be a removal/abandonment (i.e. a person standing next to the abandoned object for a period of time e.g. Video07.avi), uses a mapping across several frames, counting for how many consecutive frames the pixel has changed for. Pixels that “flicker” in changes, will cause +1 and –1 in the counter. If PERSISTENCE_THRESHOLD is reached, it is likely a real static change.
- This final persistence mapping is again cleaned with opening and closing.
- This step is not infallible, and people who stand very still for a number of frames will likely still meet the threshold.
- This step primarily filters out people who are walking very slowly or hesitate for a *short* time.

Development Stage 1 - Step 7 – Connected Components Analysis:

- CCA is used on the regions calculated by Step 6. The binary mask of the persisting changes is connected to reveal tangible objects.

Development Stage 1 - Step 8 – Draw Bounding Boxes for Persistent Changes:

- Green boundary boxes are drawn around the CCA objects from Step 7, declaring something that changes and remains changed.

Development Stage 2 - Step 9 – Abandoned vs. Removed:

- The regions of interest (defined by Step 7) are extracted and compared to the long-term background of that same region.
- These extracts are converted to greyscale, and their mean intensity difference of the greyscale is calculated.
- If the region gets significantly darker, an object is assumed added.
- If the region gets significantly lighter, an object is assumed removed.
- If the region has no major change, the change may be a result of lighting changes or other subtle changes.
- Classification labels, as well as differing colours are added for readability.

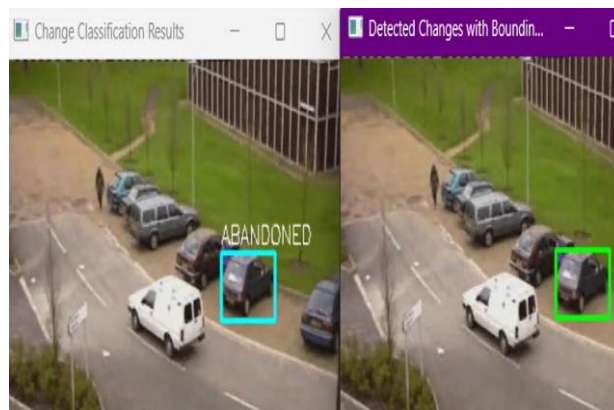


Fig 2.2 – RESULT: Successful Detection of a Removed & Abandoned Object

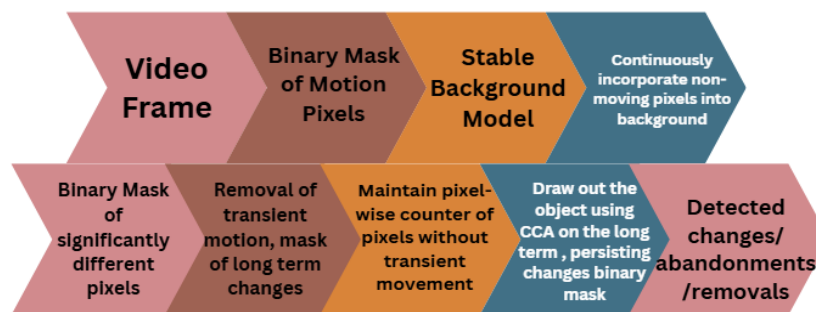


Fig 1.3 – DEVELOPMENT: Process Flow

- The above solution is relatively computationally complex, using each of these steps per-pixel. This is an intentional design choice to allow the system to be used on a generic basis.
- The per-pixel approach allows flexibility; small object abandonments or removals are often captured well.
- It also allows for subtle lighting changes throughout the day, by basing the system around a GMM background, beneficial for a real-world surveillance setting.

- The use of greyscale intensity differencing to detect abandonments/removals means there is no reliance on the size or shape of the object in question. This makes the system suitable for generic application.
- The above solution may be ineffective in nighttime situations where there is minimal difference between the dark background and the object.
- It may also be ineffective in scenarios where objects are abandoned and then changed by external forces e.g. someone dumping rubbish, then the rubbish is blown by the wind, as very slow or subtle changes are lost due to the parameter controlling how much movement needs to occur to be considered changed.

Solution Shortcomings:

- Detection *speed* and *noise* were the main issues of the system. Different approaches were attempted to mitigate the slowness of detection, e.g. combining persistent changes with a bitwise OR of the short-term mask. This negatively impacted the accuracy metrics of the system.
- More accurate detections of the objects were coupled with additional noise. Less noisy systems struggled to accurately locate the object in question.
- I have commented out the bitwise OR workaround to preserve the visual cleanliness of the system. I feel that this is the best decision which better reflects surveillance systems in a real-world scenario; additional noise is not desirable for a person who is viewing long-term change detections.

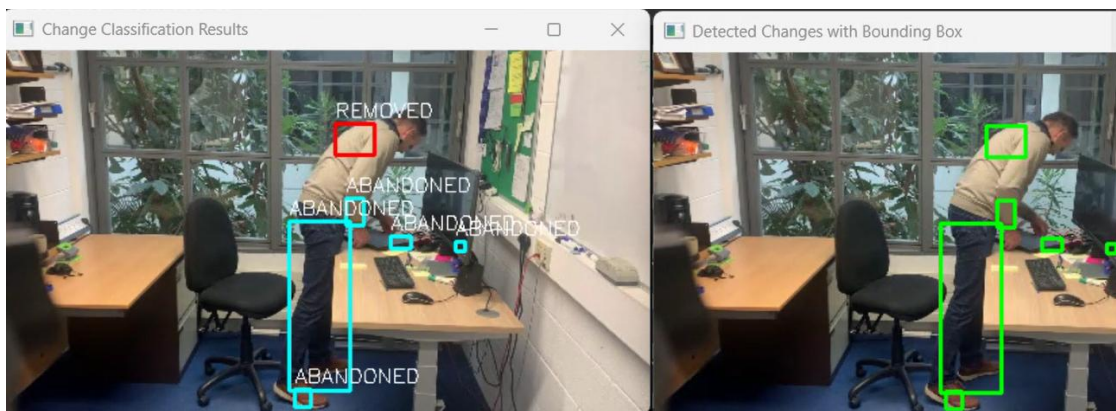


Fig 2.3 – RESULT: A Person Standing in Place for a Long Time

- Another system shortcoming is its failure to identify *other changes* which are not abandonments or removals. This is due to the simplistic nature of how abandonments and removals are classified as the intensity of change of light. The system treats moving objects as being “abandoned” into the space they now inhabit and “removed” from the space which they have stopped occupying.

- This is a limitation of how the ground truth declares the notions of abandonment and removal. Pedantically, it could be argued that a chair (Video10.avi) being moved from one location to another is “removed” from the old location and “abandoned” into the new.

Evaluation Metrics:

- As discussed above, the solution introduces a delay in detection, which originally resulted in all metrics tested evaluating to zero.
- To mitigate this, and give a more accurate reflection of the system’s accuracy, I have offset the frame count in my evaluation by the sum of variables which introduce delay, i.e. where the system waits for a period of time for certainty.
- This offset is applied to each frame where a change is detected e.g. if a change is detected in frame 90, the frame evaluated against the ground truth is 90 – FRAME_ADJUSTMENT.
- This offset allows evaluations to better reflect the detection capabilities of the system.
- This improves metrics on several videos, but it is generically applied, therefore some evaluations remain zero.
- I have chosen to evaluate the system over its:
 - Precision (of all detected changes, how many were correct)
 - Recall (of all true changes, how many were detected)
 - F1 Score (Mean of precision and recall)
 - IoU (how well the detected region overlaps the region specified in the ground truth).
- These are all reasonable These are all reasonable metrics for a surveillance system to evaluate, as a surveillance system (similar to the static camera setup of this application) must ensure that the region it is analysing is accurate (IoU) and it can detect changes accurately (precision) and reliably (recall)

Video	Ground Truth	Precision	Recall	F1	IoU
1	115, REMOVED	0.349124	0.265873	0.301864	0.177762
2	87, ABANDONED	0.130745	0.74578	0.222486	0.125167
3	87, REMOVED	0.0566285	0.0588435	0.0577148	0.0297149
4	91, ABANDONED	1	0.341153	0.508746	0.341153
4	255, REMOVED	1	0.309873	0.473134	0.309873
5	137, ABANDONED	0	0	0	0
5	357, REMOVED	0.208333	0.37037	0.266667	0.153846
6	127, ABANDONED	0.195946	0.123932	0.151832	0.082153
6	379, REMOVED	0.407801	0.491453	0.445736	0.286783
7	555, ABANDONED	0	0	0	0
8	333, ABANDONED	0	0	0	0
9	331, ABANDONED	1	0.0948485	0.173263	0.0948485
10	73, OTHER	0.304438	0.109068	0.1606	0.0873111
10	83, OTHER	0.0747207	0.153405	0.100493	0.0529048

- These metrics show a moderately accurate system, which has generically-applied evaluations.
- Removals achieve a higher precision on average than abandonments.
- Abandonments also tend to achieve a higher recall than precision, with the lower precision indicating that lighting changes which are falsely detected lean more similar to an abandonment than to a removal.
- I have also chosen to provide a table detailing visual classification (bounding boxes) against if the change was captured automatically by the ground truth. As there are several cases where metrics failed but a clearly reasonable bounding box and accurate classification can be seen by the viewer.

Video	Ground Truth	Visual Classification	Evaluation (Automatic) Classification
1	115, REMOVED	REMOVED	REMOVED
2	87, ABANDONED	ABANDONED	ABANDONED
3	87, REMOVED	REMOVED	–
4	91, ABANDONED	ABANDONED	ABANDONED
4	255, REMOVED	REMOVED	REMOVED
5	137, ABANDONED	–	–
5	357, REMOVED	REMOVED	REMOVED
6	127, ABANDONED	ABANDONED	–
6	379, REMOVED	REMOVED	REMOVED
7	555, ABANDONED	ABANDONED	–
8	333, ABANDONED	ABANDONED	–
9	331, ABANDONED	ABANDONED	ABANDONED
10	73, OTHER	ABANDONED	ABANDONED
10	83, OTHER	ABANDONED	ABANDONED

- As seen in the above table, some detections visually display a classification, but this is not achieved in a fast enough time before the ground truth evaluations detect it. E.g. Video07.avi:

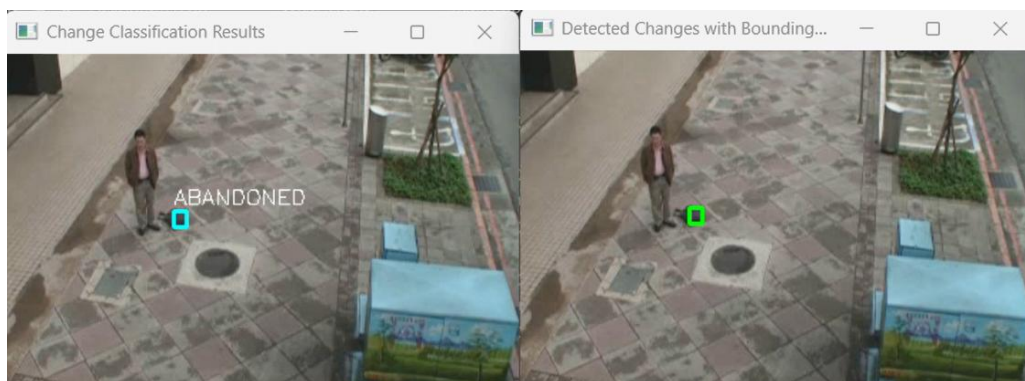


Fig 2.4 – RESULT: Visual Classification on a Video where Automatic Classification Failed

- Another result of the system is the noise resulting from people standing in place for a long time. Step 6 attempts to mitigate this and is successful in several videos (Video06.avi and Video08.avi notably). People standing in place for a very long time will trigger the threshold in Step 6. e.g. Video09.avi:
- Despite this drawback, the object in question remains reliably classified.

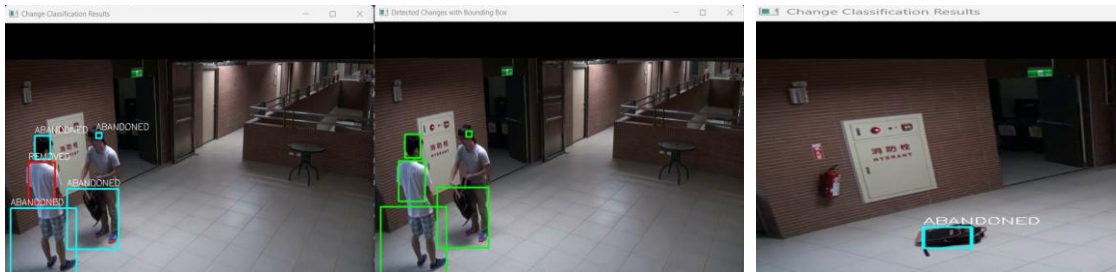


Fig 2.5 – RESULT: A Correctly Identified Abandonment Despite False Positives Overall

Possible Enhancements:

Better Tracking:

- Video feature tracking could have been implemented to better track the transient motion in the scene and subsequently remove it from classification.
- Optical Flow techniques could be implemented quite easily in the future, as the general region of interest is successfully detected by the system.
- The reason that Optical Flow was not implemented to begin with was my unfamiliarity with the topic at the stage of the course when the assignment was released.

Better Segmentation:

- A notable quirk of this solution is how objects appear fragmented in detection. Even people who are false positives appear fragmented, with their head, torso and legs all having different bounding boxes (Fig 2.5)
- This could have been corrected with a more robust edge detection implementation over Connected Components Analysis. The placement of the CCA in the implementation occurs after several thresholding operations have been carried out. Therefore, gaps in the binary videos will produce distinct labels in CCA. Applying another edge detection technique such as Robert's edge detection *before* thresholding could have prevented the disruption in the bounding boxes.
- This was not implemented because my focus was on identifying regions of interest as a priority. Edge detection before any regions of interest have been identified means that all edges in all frames would have to be analysed before actually finding regions relevant to this application.

Conclusion:

Overall, I am satisfied with the solution I have presented, regardless of drawbacks.

I feel that this solution can be reasonably applied to real-world surveillance applications.

The solution is relatively reliable, robust and scalable, with several different enhancements which could be implemented without total refactoring of the code.